

14. Why does byte code make Java a portable language?
15. Is encapsulation a characteristic of procedural or object-oriented programming?
16. Why should an object hide its data?
17. What part of an object forms an interface through which outside code may access the object's data?
18. What type of program do you use to write Java source code?
19. Will the Java compiler translate a source file that contains syntax errors?
20. What does the Java compiler translate Java source code to?
21. Assuming you are using the JDK, what command would you type at the operating system command prompt to compile the program `LabAssignment.java`?
22. Assuming there are no syntax errors in the `LabAssignment.java` program when it is compiled, answer the following questions.
 - a. What file will be produced?
 - b. What will the file contain?
 - c. What command would you type at the operating system command prompt to run the program?

Programming Challenge

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1. Your First Java Program

This assignment will help you get acquainted with your Java development software. Here is the Java program you will enter:



VideoNote

Your First Java
Program

```
// This is my first Java program.
public class MyFirstProgram
{
    public static void main(String[] args)
    {
        System.out.println("Hello World!");
    }
}
```

If You Are Using the JDK at the Command Prompt:

1. Use a text editor to type the source code exactly as it is shown. Be sure to place all the punctuation characters and be careful to match the case of the letters as they are shown. Save it to a file named `MyFirstProgram.java`.
2. After saving the program, go to your operating system's command prompt and change your current directory or folder to the one that contains the Java program you just created. Then use the following command to compile the program:

```
javac MyFirstProgram.java
```